

Rean Clive Fernandes

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 Blogposts

PROFILE

As a dedicated ML engineer with a foundation in Electronics and Control Systems, I've transitioned into the field of NLP and LLMs, completing a master's thesis on fine-tuning Open LLMs for legal applications. My diverse experience not only demonstrates my rapid learning and application capabilities but also my proficiency in a broad range of concepts from optimization and control theory. This unique blend of knowledge enables me to tackle various tasks, from robotic control to language modelling, using deep learning methods.

PROFESSIONAL EXPERIENCE

2024 Oct – present
Freiburg im Breisgau,
DE

Research Associate : Legal Large Language Models

Albert-Ludwigs-Universität

Research on LLMs in Legal Applications

- **Supervised Fine-tuning:** Fine-tuning and evaluation on legal examinations.
- **Dataset curation for Alignment:** Curating preference datasets to implement DPO and other alignment techniques.
- **Decoding methods:** Working on various decoding methods to improve legal question answering at test time.

2023 Oct – 2024 Sep
Freiburg im Breisgau,
DE

Tutor : Foundations of Deep Learning

Albert-Ludwigs-Universität

- Guided students through exercises and assignments, offering support as needed.
- Helped clarify and explain complex course materials and deep learning concepts.
- Assisted in one-on-one and group sessions to improve students' understanding.
- Assisted with course administration and instructional tasks.

EDUCATION

2021 Apr – 2024 Sep
Freiburg im Breisgau,
DE

M.Sc. in Embedded Engineering, Specialization in Artificial Intelligence | GPA 2.3 | 120 ECTS

Albert-Ludwigs-Universität 

- Foundations in AI and Computer Vision (2021): Broad exposure to core ML concepts
- Advanced Control Systems and Optimization (2022): Specialized in numerical methods and robotics through miniprojects
- Reinforcement Learning and Deep Learning Focus (2023): Excelled in practical ML applications with lab courses
- LLM Specialization and Legal AI Research (2024): Culminated in outstanding master thesis

2016 Jul – 2020 Jul
Karnataka, India

B.E. Electronics and Communication Engineering | GPA 8.65 (1.67 in German Grading system) | 180 ECTS

NMAM Institute of Technology 

SKILLS

Python

Focus on Machine learning related tools

Pandas / Matplotlib / Numpy

Dataset curation, analysis and manipulation.

High Performance Computing

Familiarity HPC :

- Distributed Training and inference of LLMs
- SLURM job scheduling and resource allocation for running LLMs on remote compute clusters.

Pytorch (and Pytorch Lightning)

Experience with various Deep Learning frameworks with accelerated training:

- Variational Auto-encoders
- Deep Reinforcement Learning
- Diffusion Models

LLM Fine-tuning and Evaluation Pipelines.

End-to-end experience with LLM finetuning starting from dataset gathering to evaluation strategies.

M.SC. PROJECTS

2024 Jan – 2024 Sep

Supervised Fine-Tuning of Open LLMs for Law: Training and Evaluation for Performance in Legal Examinations

Master Thesis | Grade : 1.0 ('sehr gut') | 30 ECTS

Outcomes :

- Fine-tuned LLaMA 2 language model on legal question answering, adapting AI for domain-specific tasks.
- Curated and distilled a dataset of bar exam questions and explanations for effective model training.
- Conducted experiments to evaluate the impact of various configurations on model performance.
- Achieved 3-4x accuracy improvements over baselines, with some models meeting California MBE passing threshold.
- Laid groundwork for future research on adapting language models for complex legal reasoning tasks.

Skills : *LLM Fine-tuning | Data preprocessing and curation | LLM pipeline creation | Quantization techniques | Prompt engineering |*

2023 Jun – 2023 Aug

Implementation Of SKIMO (SKill-based MOdel-based) RL for Robotic Manipulation

Grade : 1.3 ('sehr gut') | 6 ECTS

- Used Variational Auto-encoders to learn unsupervised representations of simulated Robot Hand manipulation model.
- Using the learned representations, applied Deep Reinforcement learning to train the simulator based Robot hand to perform diverse tasks from the learnt skills

Skills: *Unsupervised Learning | Variational Autoencoders (VAEs) | Model-based Reinforcement Learning | Deep Reinforcement Learning*

2022 Oct – 2023 Feb

Comparison of Direct collocation and Direct multiple shooting based Non-linear MPC of Crazyflie 2.1

Grade : 2.3 ('gut') | 6 ECTS

- Investigated the performance of a non-linear model predictive controller (NMPC) for the Crazyflie 2.1 nano quadcopter using direct multiple shooting and direct collocation methods.
- Implemented and evaluated the controller's open loop and closed loop performance in two key tasks: hover stability and set-point navigation.
- Analyzed results to compare the effectiveness of both direct methods in solving the subsequent non-linear programming (NLP) problem after time-domain discretization.

Skills: *Optimal Control Theory | Non-Linear MPC | Control System Design | Numerical Optimisation*

2022 Nov – 2023 Feb

Analysis of Trust Region sub-problems for optimisation

Grade : 1.3 ('sehr gut') / 3 ECTS

- Developed and implemented Trust Region methods (Cauchy point, Dog-leg, Conjugate Gradient) to solve optimization problems iteratively, aiming for both local and global minima.
- Compared the methods based on their run-time and number of iterations to evaluate efficiency in reaching the optimal solution.
- Analyzed convergence rates by comparing simulation results with theoretical values to understand the accuracy and performances.

Skills : **Numerical optimisation | Mathematical modelling**

2022 Jul – 2022 Oct

Optimal Control of a Simulated Furuta Pendulum

Grade: 1.3 ('sehr gut') / 3 ECTS

- Implemented a Model Predictive Control (MPC) approach for the Furuta Pendulum using Direct Multiple Shooting, Direct Collocation, and Real-Time Iteration methods.
- Developed and tested these methods to optimize control performance, ensuring stability and precision in the pendulum's motion.
- Analyzed the computational efficiency and accuracy of each approach in solving the underlying control problem for the dynamic system.

Skills: **Numerical Simulation methods | Real-time Iterative control | MATLAB**

RESEARCH INTERESTS

Decoding methods in LLMs

A central hypothesis of my thesis was that fine-tuning involved teaching the LLM to access knowledge by introducing specific patterns of responding, thus unlocking their further potential. This idea naturally extended to the decoding strategies, which are paramount to extracting proper path of response generation. I had so far only worked with temperature based sampling methods, but the current iteration of my work beyond my thesis involves investigating different decoding methods to see what impacts the decoding strategy has on the performance, and if we infact need elaborate fine-tuning strategies to elicit this or if just tweaking decoding strategies during inference leads to a better performance.

LLM as a Judge

Human expertise is always 'the' gold standard for data. The advent of llms has made it seem more plausible that one can leverage their knowledge to augment, if not fully replace human annotators in order to collect preference data for further training and fine-tuning of model. This is all well and good, but my gut feeling says that using a black box to solve a problem for another black box opens up an ugly can of worms where we first need to ensure that the 'Judge' is actually correct. I want to learn more about this and just how reliable using an LLM as a judge could be, as the scalability of an LLM judge may be extremely cost efficient as compared to a human expert.

Mechanistic Interpretability

Following Anthropic's release of 'Golden gate' Claude, I was enamored with the idea of understand and possibly guiding LLM outputs based on the method of Dictionary Learning and this opened me up to the field of mechanistic interpretability, which I had no idea about previously. I am just beginning to dip my toes in this field more before forming an idea of what can be done, and I find Neel Nanda's Mech Interp explainer to be a good point to read about this stuff.

Reinforcement learning in Language modelling

I am interested in the study of RL in language model training and fine-tuning. My thesis in the fine-tuning of LLMs brought me across strategies to further align model responses with techniques such as RLHF, RLAIF and DPO. While DPO is currently a more preferred approach among practitioners, I am of the opinion that the RL part of RLHF is not exactly based on the paradigm of reinforcement learning and there are still interesting things to explore in the field. In my free time I read Rich Sutton's book and try out exercises to further my knowledge of RL and gain more insight into bridging the knowledge of what i have done so far with what I intend to do in my research.

LANGUAGES

English
Native proficiency

German
A2